

Pramo Jewels

System Architecture Document

Document ID: PJ-ARCH-001
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Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define the overall software architecture and major system components
Audience	Solution Architects, Developers, DevOps, Security, AI Coding Agents
Version	1.0

1. Architecture Vision

Build a secure, modular and scalable architecture capable of supporting premium shopping experiences, business growth and future mobile applications.

2. High-Level Architecture

Client Layer: Next.js (React), Tailwind CSS, Three.js / React Three Fiber.
API Layer: Laravel REST APIs.
Business Layer: Authentication, Catalogue, Orders, Payments, Shipping, Notifications.
Data Layer: MySQL, Redis (cache/session), Object Storage for media.
External Services: Payment Gateway, Courier APIs, Email, SMS/OTP, WhatsApp.

3. Core Modules

Authentication & Accounts
Catalogue & Search
Cart & Wishlist
Checkout & Payments
Orders & Returns
Inventory
Reviews
Admin Portal
Notifications
Analytics

4. Data Flow

Users interact with the frontend, which communicates with secured REST APIs. Business services validate requests, access the database or cache, integrate with third-party services, and return structured JSON responses.

5. Security Architecture

HTTPS everywhere
JWT/Sanctum authentication
RBAC
CSRF/XSS/SQL Injection protection
Rate limiting
Encrypted secrets
Audit logging

6. Scalability

Stateless application servers
CDN for assets
Redis caching
Horizontal scaling
Background queues for emails, notifications and media processing

7. Availability & Reliability

99.9% target uptime
Health checks
Automated backups
Graceful error handling
Disaster recovery procedures

8. Deployment Architecture

Production:
CDN → Reverse Proxy → Next.js Frontend
→ Laravel API
→ MySQL + Redis
→ Object Storage
→ External Integrations

9. Monitoring

Centralized logging, application monitoring, uptime monitoring, performance metrics and security alerts.

10. Future Expansion

Support mobile apps, AI recommendations, AR try-on, multilingual content, loyalty program and microservice decomposition if future scale requires.